The actual 160-bit curve has exactly p points

Lean proves the entire corrected curve has p rational points, using its certified order-p point, a closed quadratic-nonresidue certificate and an elementary point-count bound. No Hasse or CM point-count theorem is assumed. The curve and prime are the same actual Mathlib objects as the parent point and concrete Smart arithmetic certificates. [pcc:unverified]

```
p = 730750818665451459112596905638433048232067471723
a = 425706413842211054102700238164133538302169176474
b = 203362936548826936673264444982866339953265530166
E: y² = x³ + ax + b over ZMod p
```

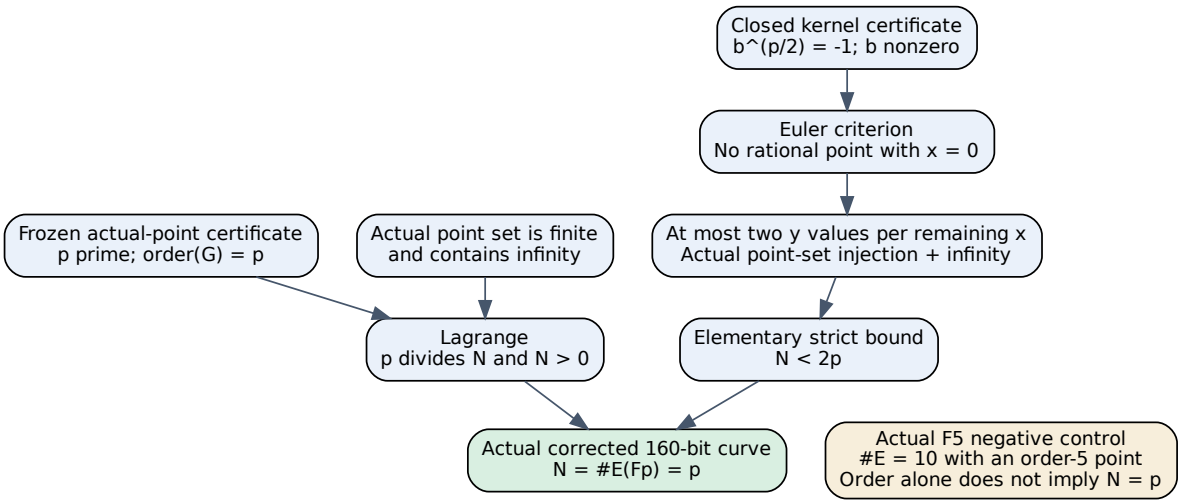


Figure 1: Elementary actual-point counting and explicit negative control. [pcc:unverified]

The kernel checks b^(p/2)=−1 and b≠0. Euler's criterion excludes x=0. At most two y-values occur at every other x, so the actual point count N satisfies N<2p. The certified point gives p divides N; finiteness and infinity give N>0. Hence N=p. [pcc:unverified]

Exact proof obligations and negative controls

Module	Checked obligation
`ShortPointCount`	Actual nonsingular-point equation; injective encoding into nonmissing x-values × Bool; strict count bound; arithmetic multiple lemma
`MissingFiber`	Closed Euler-power check, nonzero b and no point at x=0
`CardinalityControls`	Actual F_5 curve has ten points, an order-five point and no missing x-fibre; direct universal negation
`AnomalousCardinality`	Finite positive actual point count, Lagrange composition, exact N=p and combined certificate

Two actual controls prevent an unjustified identification of subgroup and curve counts. Over F_3, $y^2=x^3-x+1$ has seven points, so a universal strict $N<2q$ bound fails. Over F_5, $y^2=x^3+3x$ has ten points and an actual order-five point; all x-fibres are nonempty. Lean directly refutes the inference that an order-characteristic point alone forces total curve order equal to the characteristic. These controls do not contradict the checked missing-fibre argument. [pcc:unverified]

The generic encoding chooses one square root per nonempty fibre and records whether a point uses that root or its negative. It is a noncomputable mathematical injection, not a runtime or enumeration claim. Actual point cardinality is connected through Mathlib's nonsingular-point equivalence; no abstract lookup table or assumed curve cardinality is substituted. [pcc:unverified]

Main binding: `AnomalousCardinalityAudit.anomalous_curve_cardinality_certificate`. Full gate: 17 declarations, zero failing and zero trusted-native exceptions; generated equations are included, private proofs audited transitively. Lean 4.35.0 / Mathlib 56a00585, only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide exception, linter waiver or added elaboration setting. Failed extraction/choice/tuple proofs and the rejected modern-header setup remain archived; legacy import advisories are retained. [pcc:unverified]

Source commit `7783f692` precedes packet `PCC-20261110-541e6f`. Full local verification passes at 10:39:11Z; signed standing is unverified. Source pins, actual verdict and archive roundtrip checks are in `../../sources.json` and `../../verification.json`. This closes the former whole-curve-cardinality obligation, while the generic Smart p-adic transfer, complexity/native compiler refinement and matched SafeCurves benchmark remain open. The native execution packet remains separately `not_covered`. NIST's screened recommendation is not refuted. [pcc:unverified]

Sources checked: corrected anomalous-curve construction and author erratum; Smart HPL-97-128; Mathlib affine point, square-root, finite cardinality, Lagrange and Euler-criterion declarations. Prior paper hashes/read scopes remain in the parent plan. KB/Conductor unavailable; bounded primary, repository and Mathlib fallback. No novelty, native experiment, sampling, timing, security-cost or paid-compute claim. [pcc:unverified]